

Volume IV · Chapter 19 — Figures & Multimodal Understanding · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-IV-Chapter-19-context.md for the AI interaction log.

Prerequisites: Ch 18, Ch 7.

Learning outcomes — the student can: detect/extract figures; generate & curate captions with multimodal models; link images to text; make images retrievable; describe the SOTA and limits of medical image understanding.

Labs (hands-on) — 14 h:

#	Lab	h
19a	Extract figures from a medical document	+ 3
19b	Generate captions with a multimodal model; compare to originals	4
19c	Multimodal retrieval: query text → return relevant figures	4
19d	Link figures to text chunks; evaluate linkage quality	3

Datasets/tools: multimodal HF models (LLaVA-Med/MedGemma), CLIP-style embeddings; vector DB. **Assessment:** multimodal ingest+retrieval (**60%**); caption-

quality report (20%); quiz (20%). **Key decisions:** which multimodal model; how much to trust captions; image vs. caption embeddings. **References:** plan §6; §13 → *Medical LLMs*. **Hours:** Theory 8 + Lab 14 = 22.